

# SAVANNA SUDAN CLIMATE

SAVANNA TRANSITION MAP → ITCZ SEASONAL SWITCH → WATER-RELIABILITY CHAIN → GRASS-TREE GRADIENT → FIRE-HERBIVORY BALANCE → WILDLIFE-LIVELIHOOD MATRIX

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

## SAVANNA TRANSITION MAP

SAVANNA TRANSITION MAP

WETTER AND TALLER GRASS

SUDAN BELT

CLASSIC WET-DRY SAVANNA

SHORTER GRASS AND OPEN TREES

DESERT MARGIN

DROUGHT RISK INTENSIFIES

MUST REMEMBER

EQUATORWARD → wetter and taller grass

SUDAN BELT → classic wet-dry savanna

POLEWARD → shorter grass and open trees

DESERT MARGIN → drought risk intensifies

MUST REMEMBER: Savanna Aw/As climates reflect seasonal ITCZ migration between convective wet summers and subtropical-high

### CORE CLAIM

- EQUATORWARD → wetter and taller grass
- SUDAN BELT → classic wet-dry savanna

### MECHANISM

- POLEWARD → shorter grass and open trees
- DESERT MARGIN → drought risk intensifies

### CONSEQUENCE / LIMIT

- MUST REMEMBER: Savanna Aw/As climates reflect seasonal ITCZ migration between convective wet summers and subtropical-high dry
- winters, sustaining tall grasses with scattered drought/fire-adapted trees across Africa, South America and northern Australia.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Savanna Aw/As climates reflect seasonal ITCZ migration between convective wet summers and subtropical-high dry winters, sustaining tall grasses with scattered drought/fire-adapted trees across Africa, South America and northern Australia.

01

STAGE 01

## ITCZ SEASONAL SWITCH

ITCZ SEASONAL SWITCH

HIGH-SUN SEASON

ITCZ APPROACHES

CONVERGENCE + CONVECTION

SUMMER RAIN

LOW-SUN SEASON

DRY TRADES DOMINATE

HOT WET

HIGH-SUN SEASON → ITCZ approaches

CONVERGENCE + CONVECTION → summer rain

LOW-SUN SEASON → dry trades dominate

RESULT → hot wet / warm dry rhythm

### CORE CLAIM

- HIGH-SUN SEASON → ITCZ approaches
- CONVERGENCE + CONVECTION → summer rain

### MECHANISM

- LOW-SUN SEASON → dry trades dominate
- RESULT → hot wet / warm dry rhythm

### CONSEQUENCE / LIMIT

- HIGH-SUN SEASON → ITCZ approaches
- CONVERGENCE + CONVECTION → summer rain

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → hot wet / warm dry rhythm.

02

STAGE 02

## WATER-RELIABILITY CHAIN

WATER-RELIABILITY CHAIN

LATE ONSET

DELAYED SOWING

MID-SEASON BREAK

CROP AND FORAGE STRESS

EARLY CESSATION

INCOMPLETE GRAIN FILLING

MIGRATION AND HERD LOSS RISK

LATE ONSET → delayed sowing

MID-SEASON BREAK → crop and forage stress

EARLY CESSATION → incomplete grain filling

DROUGHT → migration and herd loss risk

### CORE CLAIM

- LATE ONSET → delayed sowing
- MID-SEASON BREAK → crop and forage stress

### MECHANISM

- EARLY CESSATION → incomplete grain filling
- DROUGHT → migration and herd loss risk

### CONSEQUENCE / LIMIT

- LATE ONSET → delayed sowing
- MID-SEASON BREAK → crop and forage stress

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARLY CESSATION → incomplete grain filling; DROUGHT → migration and herd loss risk.

03

STAGE 03

## GRASS-TREE GRADIENT

GRASS-TREE GRADIENT

WET MARGIN

TALL GRASS + MORE TREES

CORE SAVANNA

PARKLAND MOSAIC

DRY MARGIN

SHORT GRASS + THORNY SHRUBS

DESERT EDGE

WET MARGIN → tall grass + more trees

CORE SAVANNA → parkland mosaic

DRY MARGIN → short grass + thorny shrubs

DESERT EDGE → discontinuous vegetation

- LATE ONSET → delayed sowing
- MID-SEASON BREAK → crop and forage stress

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARLY CESSATION → incomplete grain filling; DROUGHT → migration and herd loss risk.**

**WET MARGIN** → tall grass + more trees

→ **DESERT EDGE** → discontinuous vegetation

- WET MARGIN → tall grass + more trees
- CORE SAVANNA → parkland mosaic

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY MARGIN → short grass + thorny shrubs.**

➡ **THICK BARK + ROOTS** → selected tree survival

- SEASONAL RAIN → grass biomass
- DRY SEASON → combustible fuel

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THICK BARK + ROOTS → selected tree survival.**

|                                            |
|--------------------------------------------|
| HERDS → follow seasonal forage and water   |
| PREDATORS → track herbivore concentrations |

- HERDS → follow seasonal forage and water
- PREDATORS → track herbivore concentrations

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PASTORALISTS → mobility spreads risk; CULTIVATORS → millet / sorghum timing.**

➔ **FIXED WATER POINT** → local grazing concentration

- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY SEASON → fire and surface sealing; FIXED WATER POINT → local grazing concentration.**

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PASTORALISTS → mobility spreads risk; CULTIVATORS → millet / sorghum timing.

06

STAGE 06 DEGRADATION PATHWAY

DEGRADATION PATHWAY COVER LOSS RAINDROP AND WIND EXPOSURE WET SEASON RUNOFF, LEACHING AND EROSION DRY SEASON FIRE AND SURFACE SEALING FIXED WATER POINT

COVER LOSS → raindrop and wind exposure

WET SEASON → runoff, leaching and erosion

DRY SEASON → fire and surface sealing

FIXED WATER POINT → local grazing concentration

CORE CLAIM

- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

MECHANISM

- DRY SEASON → fire and surface sealing
- FIXED WATER POINT → local grazing concentration

CONSEQUENCE / LIMIT

- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY SEASON → fire and surface sealing; FIXED WATER POINT → local grazing concentration.

07

STAGE 07 INDIA FOREST GRADIENT

INDIA FOREST GRADIENT MOIST DECIDUOUS LONGER WET SEASON DRY DECIDUOUS STRONGER LEAF-SHEDDING SEMI-ARID MOISTURE STRESS OVERLAPPING BELTS, NOT FIXED WALLS

CORE CLAIM

MOIST DECIDUOUS → longer wet season  
DRY DECIDUOUS → stronger leaf-shedding

MECHANISM

THORN / SCRUB → semi-arid moisture stress  
RULE → overlapping belts, not fixed walls

CONSEQUENCE / LIMIT

MOIST DECIDUOUS → longer wet season  
DRY DECIDUOUS → stronger leaf-shedding

CORE CLAIM

- MOIST DECIDUOUS → longer wet season
- DRY DECIDUOUS → stronger leaf-shedding

MECHANISM

- THORN / SCRUB → semi-arid moisture stress
- RULE → overlapping belts, not fixed walls

CONSEQUENCE / LIMIT

- MOIST DECIDUOUS → longer wet season
- DRY DECIDUOUS → stronger leaf-shedding

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THORN / SCRUB → semi-arid moisture stress; RULE → overlapping belts, not fixed walls.

08

STAGE 08 INDIAN GRASSLAND FIREWALL

INDIAN GRASSLAND FIREWALL HUMID TALL GRASS AND FLOODPLAIN MONTANE GRASS-FOREST MOSAIC ARID-SALINE KACHCHH GRASSLAND HYDROLOGY AND HISTORY FIRST

CORE CLAIM

TERAI → humid tall grass and floodplain  
SHOLA → montane grass-forest mosaic

MECHANISM

BANNI → arid-saline Kachchh grassland  
MANAGEMENT → hydrology and history first

CONSEQUENCE / LIMIT

TERAI → humid tall grass and floodplain  
SHOLA → montane grass-forest mosaic

CORE CLAIM

- TERAI → humid tall grass and floodplain
- SHOLA → montane grass-forest mosaic

MECHANISM

- BANNI → arid-saline Kachchh grassland
- MANAGEMENT → hydrology and history first

CONSEQUENCE / LIMIT

- TERAI → humid tall grass and floodplain
- SHOLA → montane grass-forest mosaic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BANNI → arid-saline Kachchh grassland; MANAGEMENT → hydrology and history first.

09

STAGE 09 RESTORATION DECISION TREE

RESTORATION DECISION TREE NATURAL OPEN ECOSYSTEM CONSERVE OPENNESS DEGRADED SCRUB RESTORE NATIVE MOSAIC WATER PROCESS RECHARGE AND DRAINAGE GRAZING AND TENURE DESIGN

NATURAL OPEN ECOSYSTEM → conserve openness

DEGRADED SCRUB → restore native mosaic

WATER PROCESS → recharge and drainage

LIVELIHOOD → grazing and tenure design

CORE CLAIM

- NATURAL OPEN ECOSYSTEM → conserve openness
- DEGRADED SCRUB → restore native mosaic

MECHANISM

- WATER PROCESS → recharge and drainage
- LIVELIHOOD → grazing and tenure design

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical grassland, and parkland structure is not
- evidence of a climax maintained by climate alone because fire, herbivory and land use matter.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical grassland, and parkland structure is not evidence of a climax maintained by climate alone because fire, herbivory and land use matter.

RESTORATION DECISION TREE

NATURAL OPEN ECOSYSTEM

CONSERVE OPENNESS

DEGRADED SCRUB

RESTORE NATIVE MOSAIC

WATER PROCESS

RECHARGE AND DRAINAGE

GRAZING AND TENURE DESIGN

NATURAL OPEN ECOSYSTEM → conserve openness

DEGRADED SCRUB → restore native mosaic

WATER PROCESS → recharge and drainage

LIVELIHOOD → grazing and tenure design

CORE CLAIM

NATURAL OPEN ECOSYSTEM → conserve openness

DEGRADED SCRUB → restore native mosaic

MECHANISM

WATER PROCESS → recharge and drainage

LIVELIHOOD → grazing and tenure design

CONSEQUENCE / LIMIT

CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical grassland, and parkland structure is not

evidence of a climax maintained by climate alone because fire, herbivory and land use matter.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical grassland, and parkland structure is not evidence of a climax maintained by climate alone because fire, herbivory and land use matter.

10

STAGE 10

SAVANNA PYQ FIREWALL

SAVANNA PYQ FIREWALL

2021 PRELIMS

TREE-LIMITING CONDITIONS

LONG DRY SEASON

FIRE + HERBIVORY

NO ANSWER LETTER INVENTED

EVIDENCE LIMIT

FIRE CAN RECYCLE NUTRIENTS AND MAINTAIN OPENNESS BUT FREQUENCY/INTENSITY

CORE CLAIM

2021 PRELIMS → tree-limiting conditions

CLIMATE → long dry season

MECHANISM

DISTURBANCE → fire + herbivory

LEDGER → no answer letter invented

CONSEQUENCE / LIMIT

EVIDENCE LIMIT: Fire can recycle nutrients and maintain openness but frequency/intensity changes may degrade systems

Indian deciduous forests and grasslands need region-specific rainfall, fire and grazing evidence rather than direct Sudan-type equivalence.

CORE CLAIM

2021 PRELIMS → tree-limiting conditions

CLIMATE → long dry season

MECHANISM

DISTURBANCE → fire + herbivory

LEDGER → no answer letter invented

CONSEQUENCE / LIMIT

EVIDENCE LIMIT: Fire can recycle nutrients and maintain openness but frequency/intensity changes may degrade systems

Indian deciduous forests and grasslands need region-specific rainfall, fire and grazing evidence rather than direct Sudan-type equivalence.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Fire can recycle nutrients and maintain openness but frequency/intensity changes may degrade systems; Indian deciduous forests and grasslands need region-specific rainfall, fire and grazing evidence rather than direct Sudan-type equivalence.

11

STAGE 11

SAVANNA ANSWER SPINE

SAVANNA ANSWER SPINE

RAINFOREST-DESERT TRANSITION

ITCZ AND WET-DRY SEASON

GRASS-TREE DISTURBANCE BALANCE

INDIA MOSAICS AND RESTORATION

LOCATE → rainforest-desert transition

DERIVE → ITCZ and wet-dry season

EXPLAIN → grass-tree disturbance balance

APPLY → India mosaics and restoration

CORE CLAIM

LOCATE → rainforest-desert transition

DERIVE → ITCZ and wet-dry season

MECHANISM

EXPLAIN → grass-tree disturbance balance

APPLY → India mosaics and restoration

CONSEQUENCE / LIMIT

LOCATE → rainforest-desert transition

DERIVE → ITCZ and wet-dry season

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPLAIN → grass-tree disturbance balance; APPLY → India mosaics and restoration.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

INDIA'S SAVANNA ANALOGUE

DRY DECIDUOUS FOREST + THORN SCRUB + GRASSLANDS .

TROPICAL MOIST DECIDUOUS FORESTS

100–200 CM RAINFALL; DOMINANT TEAK AND SAL .

MOIST DECIDUOUS REGIONS

SAHYADRI , NE PENINSULA, HIMALAYAN FOOTHILLS.

THORN/SCRUB REFLECTS ARIDITY PLUS LAND-USE PRESSURE; IT IS NOT

OPTIONAL SOURCE DEPTH

India's savanna analogue = dry deciduous forest + thorn scrub + grasslands .

Tropical moist deciduous forests: 100–200 cm rainfall; dominant teak and sal .

Moist deciduous regions: Sahyadris , NE peninsula, Himalayan foothills.

Thorn/scrub reflects aridity plus land-use pressure; it is not invariably degraded moist deciduous forest.

ADVANCED DEBATE

Rainfall thresholds overlap regionally; use approximate belts rather than rigid universal cut-offs.

Thorn forest regions: peninsular India, Rajasthan, Haryana, Punjab, W UP, Kachchh, MP, Himalayan foothills.

Deciduous trees shed leaves in dry season.

Grasslands include Terai, shola and Banni examples in the source notes.

LEGACY / NUANCE

Fire and grazing maintain open grassland/woodland mosaics.

NOT: Dry deciduous forests are evergreen → They are deciduous and shed leaves in the dry season.

NOT: Thorn forests occur in the wettest parts of India → They occur in arid/semi-arid belts, generally

NOT: Banni and Terai grasslands are the same type → Terai is humid tall wet grassland; Banni is arid/saline Kachchh grassland.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.